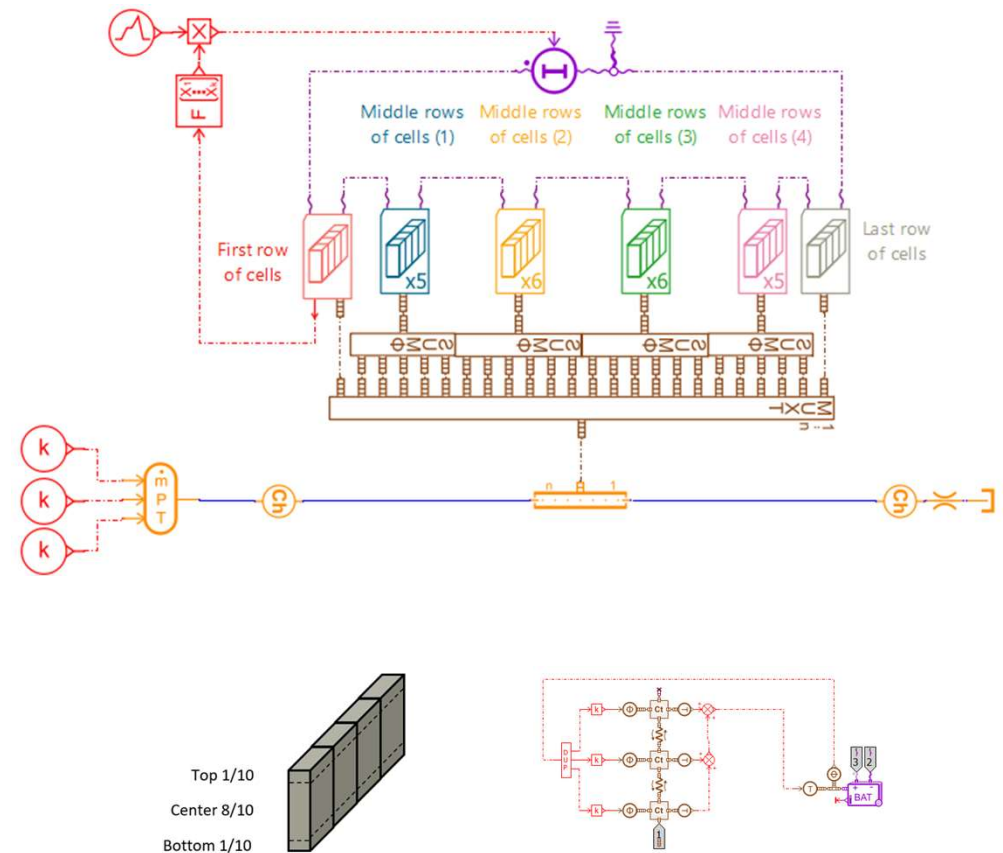
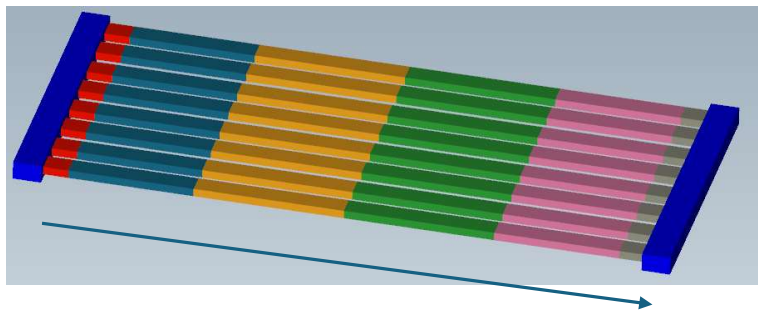
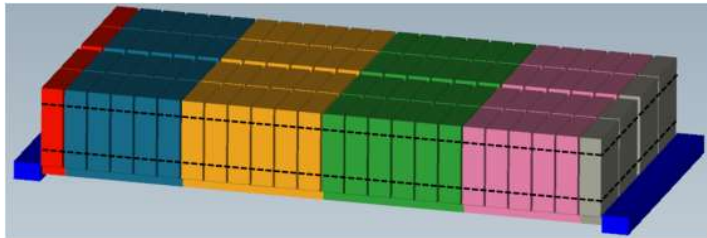


Thermal Analysis

Dongmyung Suh

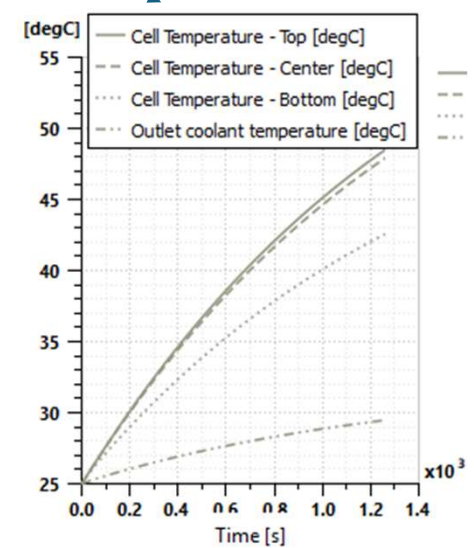
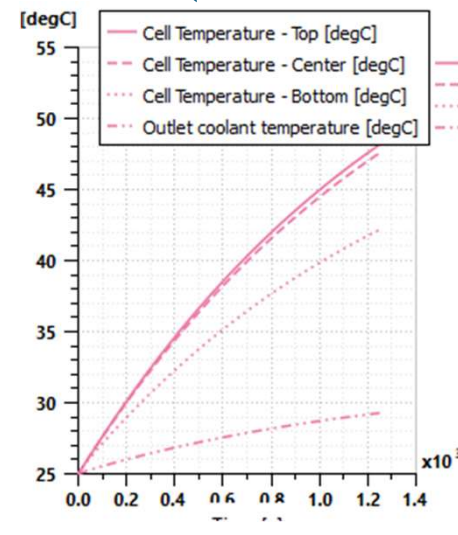
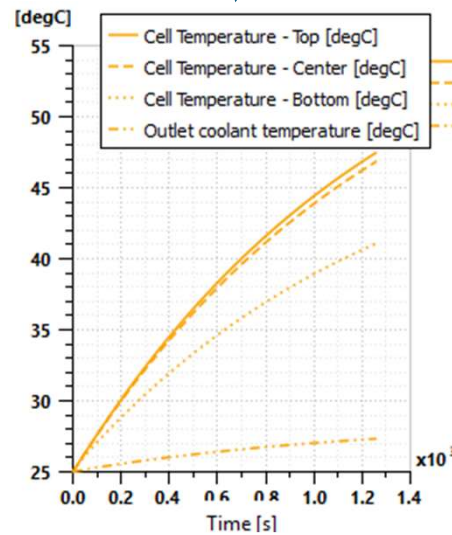
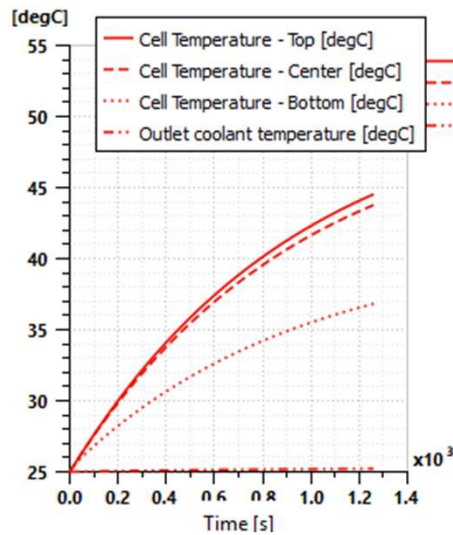
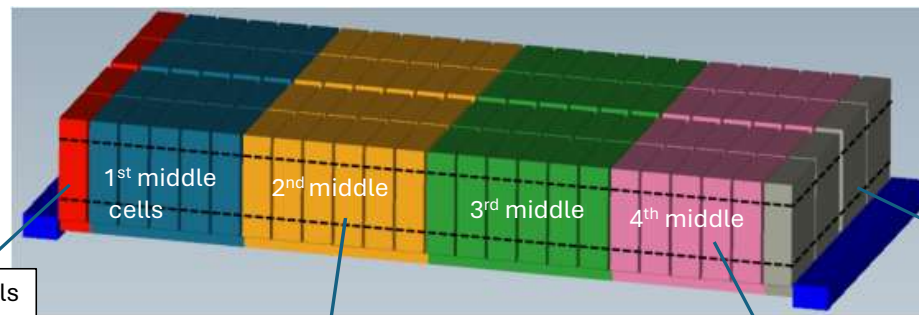
Thermal Analysis for Battery Package

Battery Package cooled by liquid (Amesim Model)



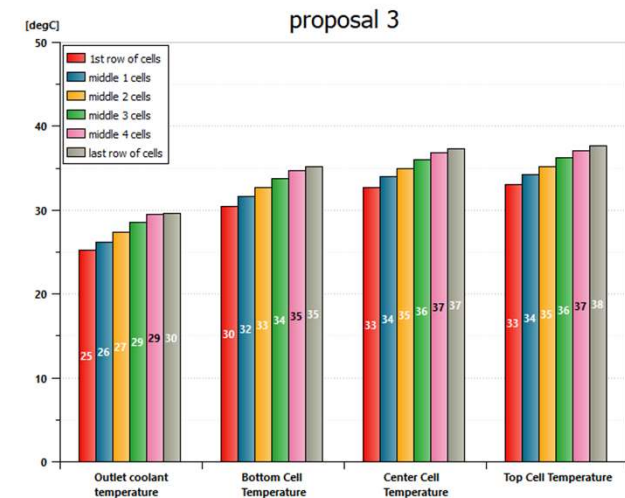
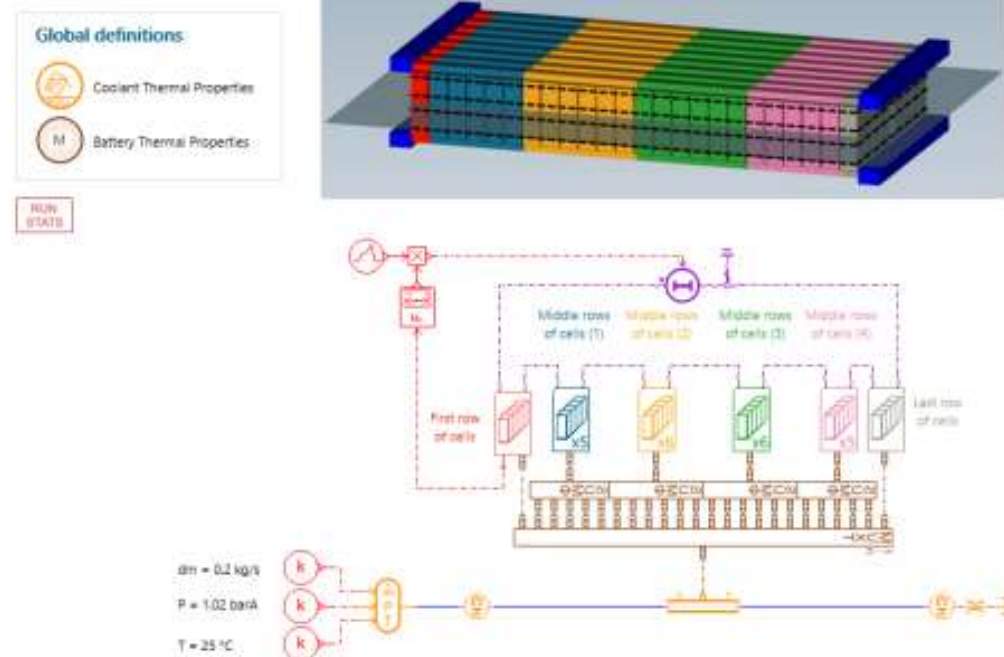
Thermal management requirements for performance, aging and safety

- Cell temperature < 40 °C
- Temperature gradient in a cell < 5 °C
- Temperature gradient between neighboring cells < 5 °C



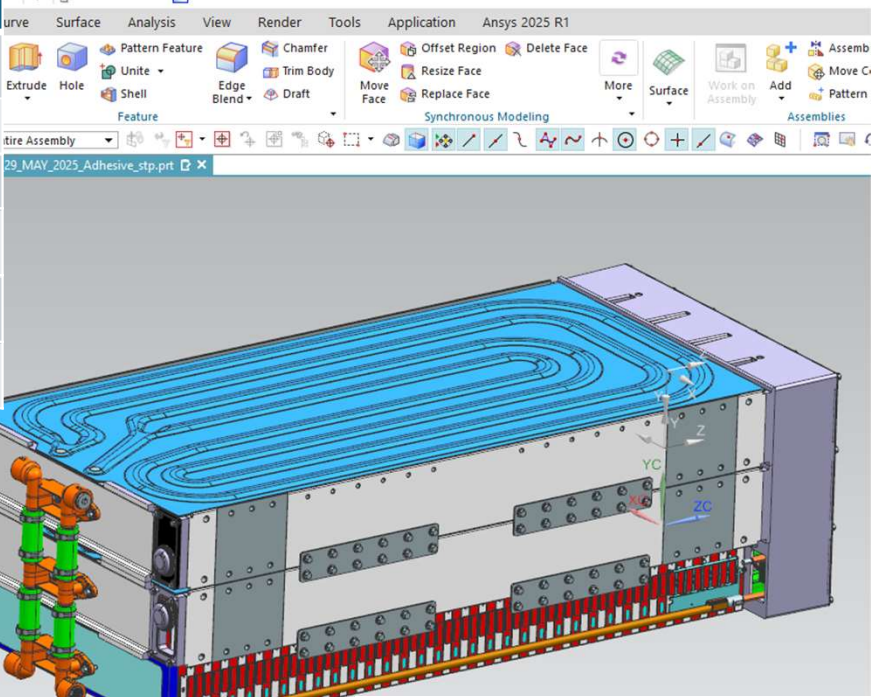
Battery Pack cooled by Coolant (Amesim)

Liquid-cooled battery pack - Step 2

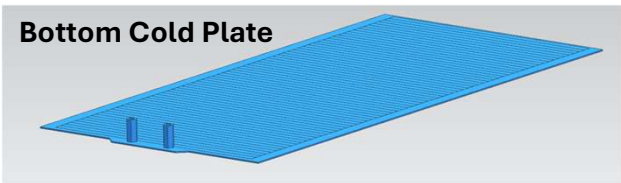
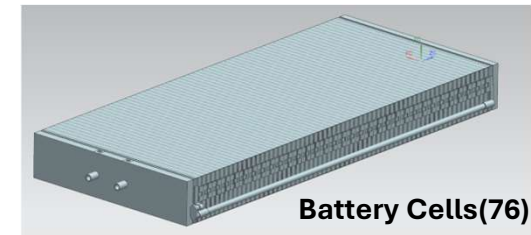
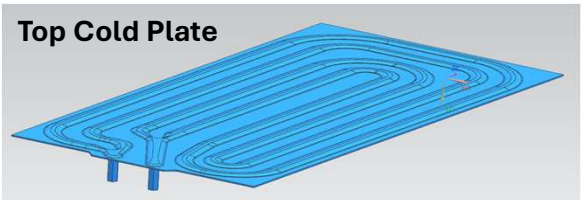


Thermal Analysis in Battery package

Parameter	BP104
Cell nominal voltage (V)	3.63
Cell Capacity (Ahr)	143
Cell Energy (Wh)	520
3 Tier Atlas Pack Cell #	200
3 Tier Atlas Pack Energy (kWh)	104
3 Tier Atlas Pack nominal voltage (V)	726

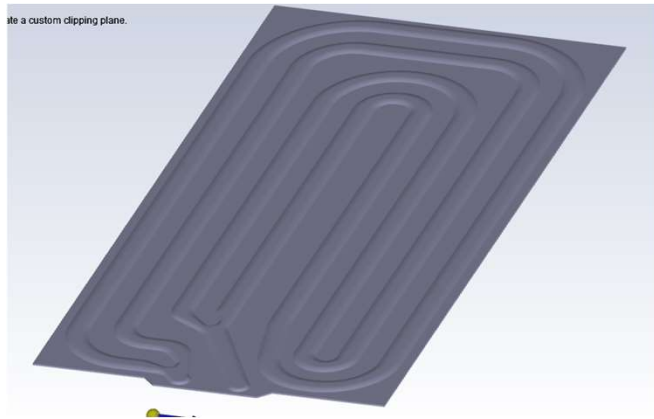


3Tier Battery Package (BP104)
Total # of components: 2250

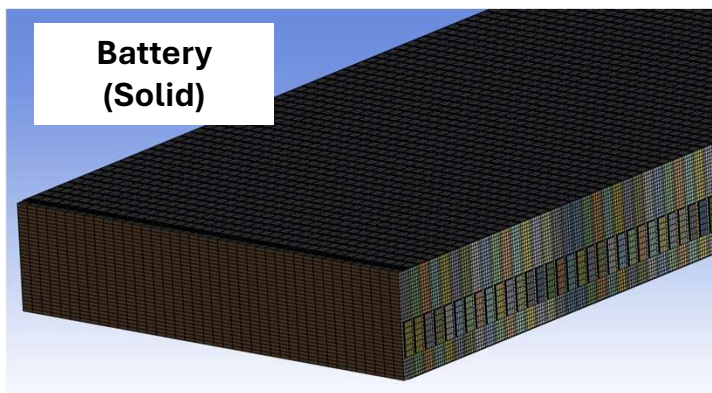


Clean-up & Simplifying of Geometry Model (Siemens NX)

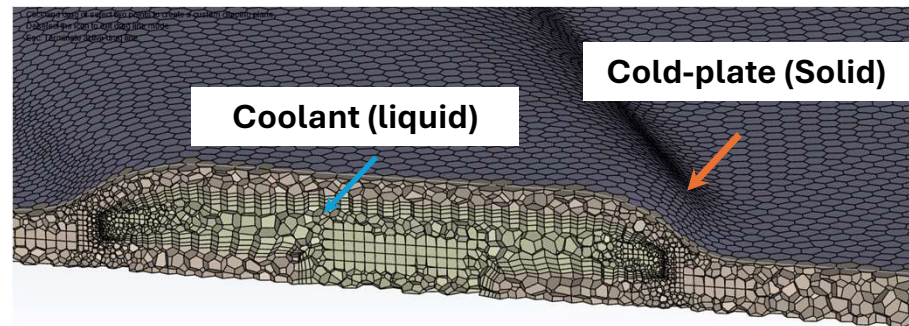
Meshing



Ansys meshing (Hexagonal mesh)

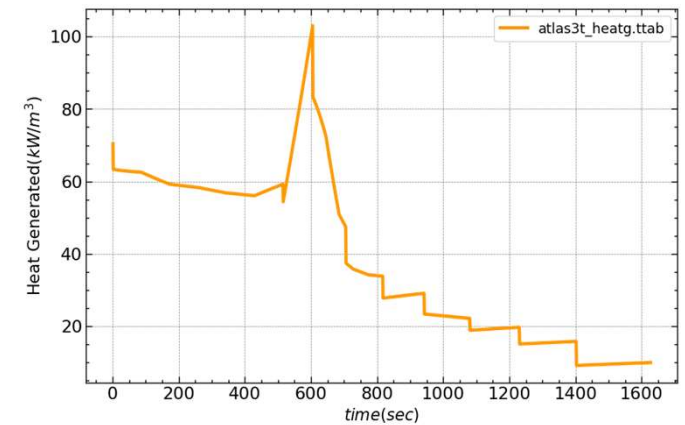
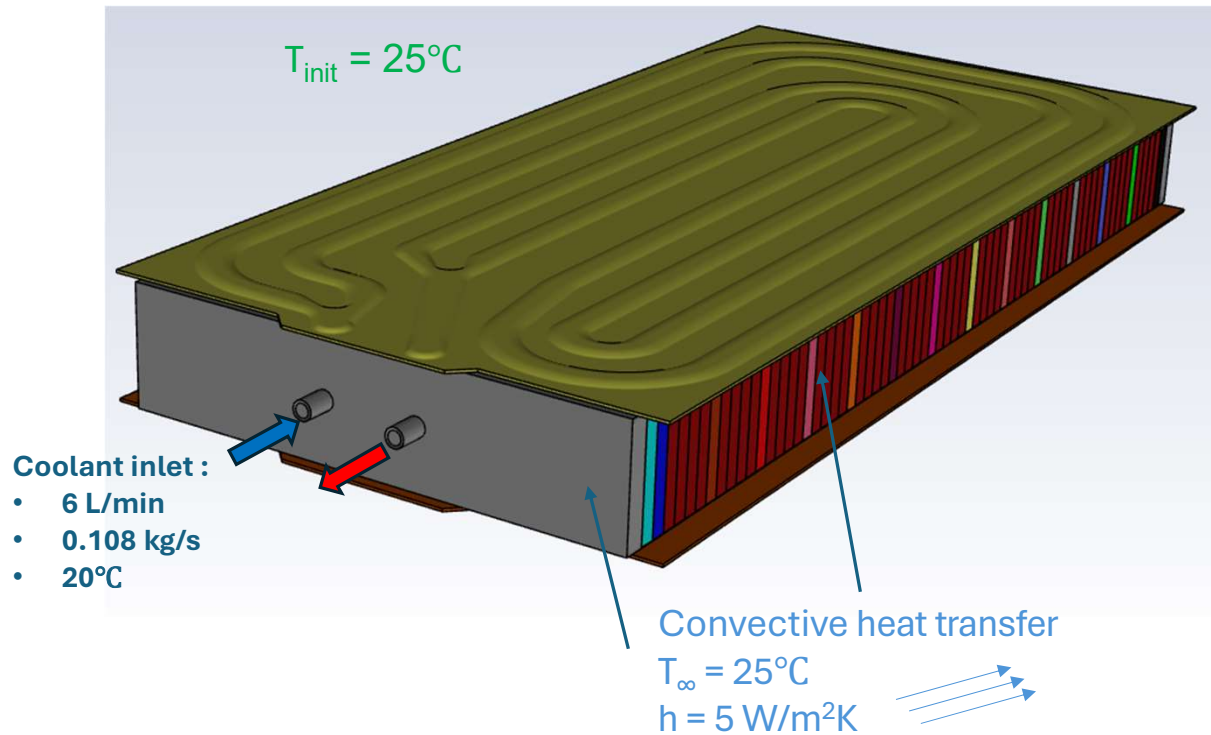


Ansys Watertight meshing (polyhedral mesh)



- **Conjugated Heat Transfer between Cold-plate and Coolant**
- **Total 20 million meshes**

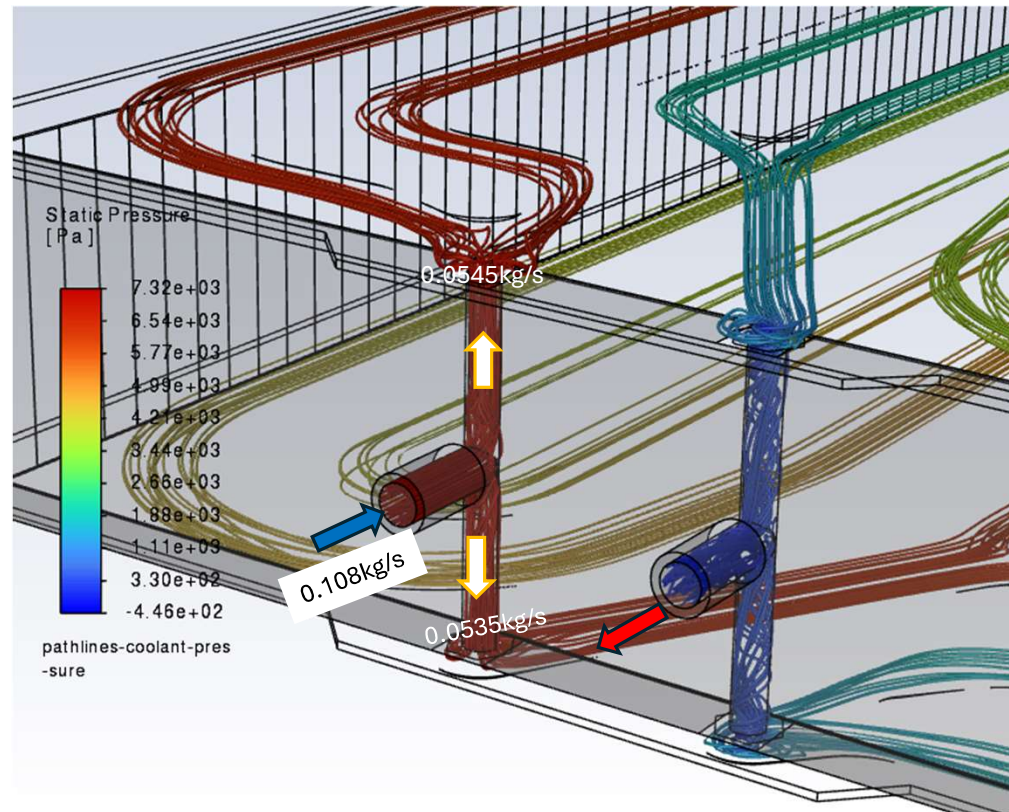
Boundary Condition & Heat Source



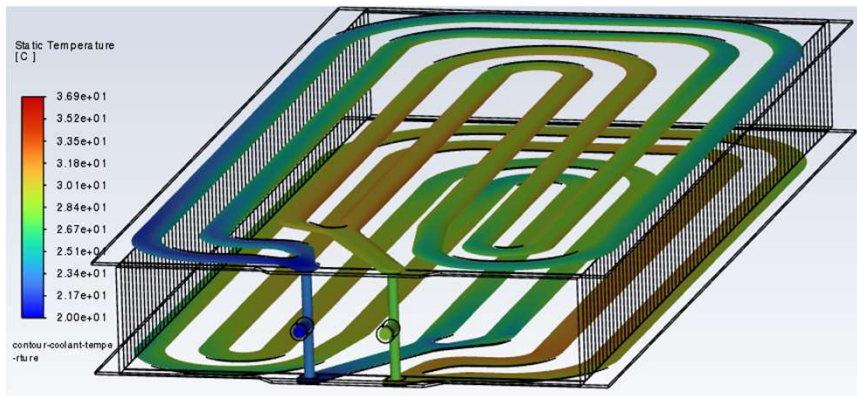
Heat generation while fast charging cells

Coolant Flow (mass flow rate and pressure drop)

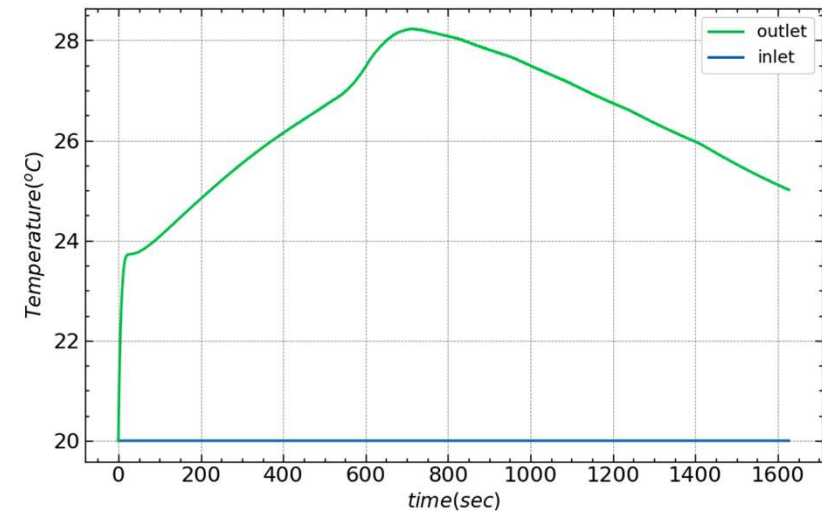
$$P(\text{inlet}) - p(\text{outlet}) = \Delta P = 7.32 \text{ kPa}$$



3T – Temperature evolution of Coolant

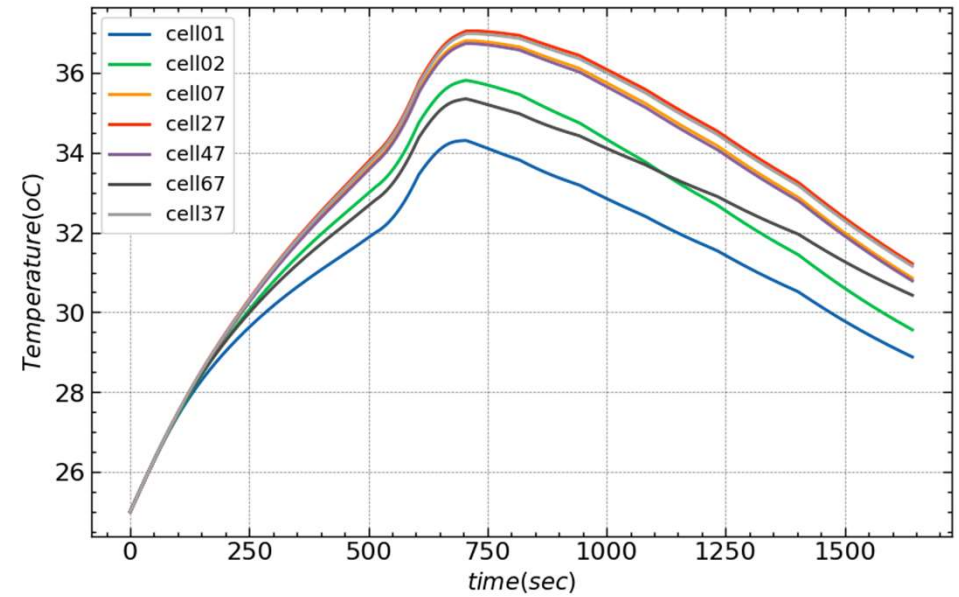
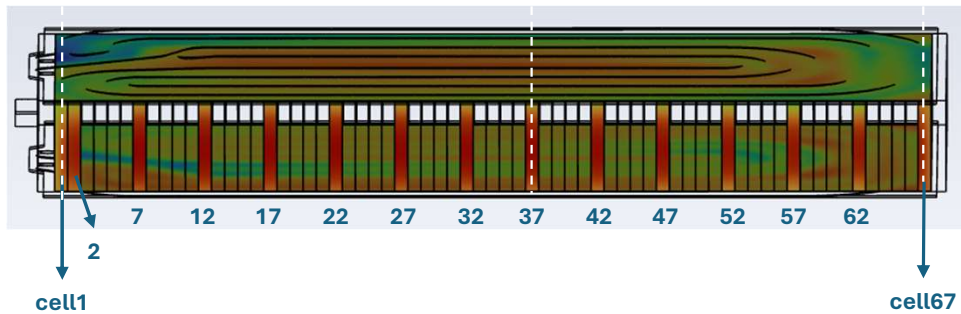


Coolant Temperature in flow channels @800s

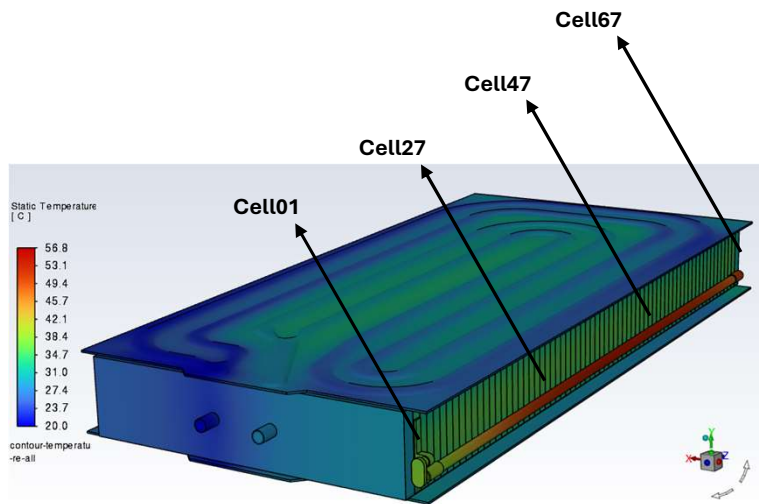


Coolant Temperature Evolution

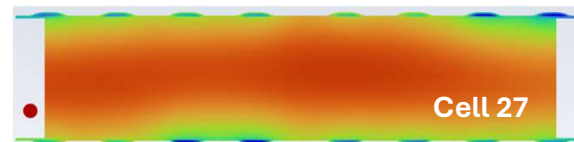
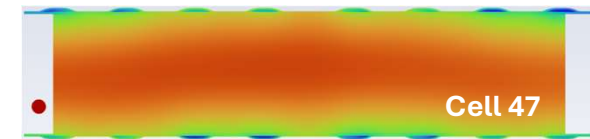
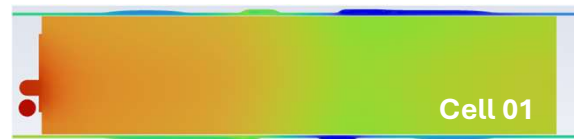
Temperature evolution of Cells



Cell temperature at mid vertical section



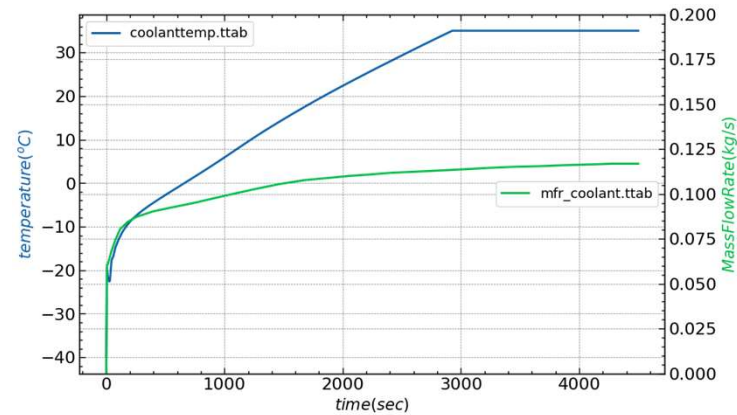
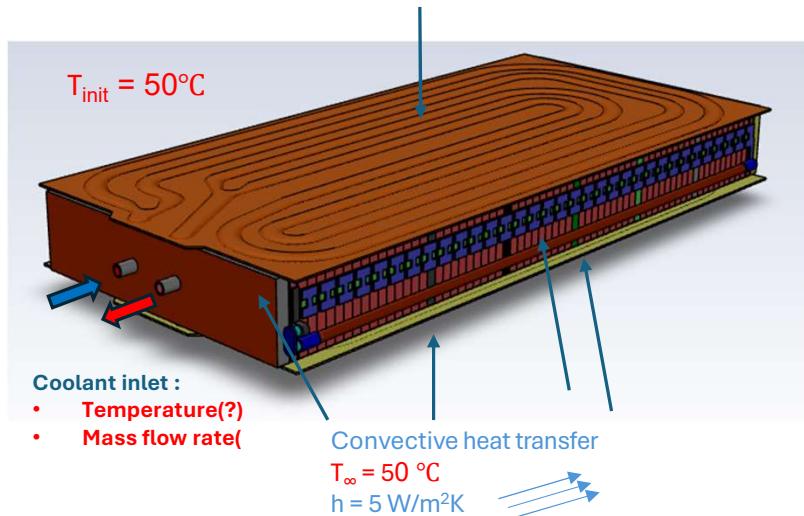
Thermal Analysis: Temperature distribution



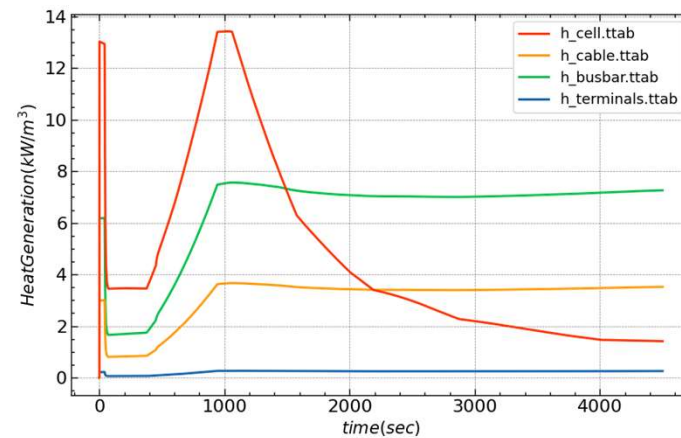
Temperature Distribution of Cells

COLD Start (-40 °C)

Simulation Condition

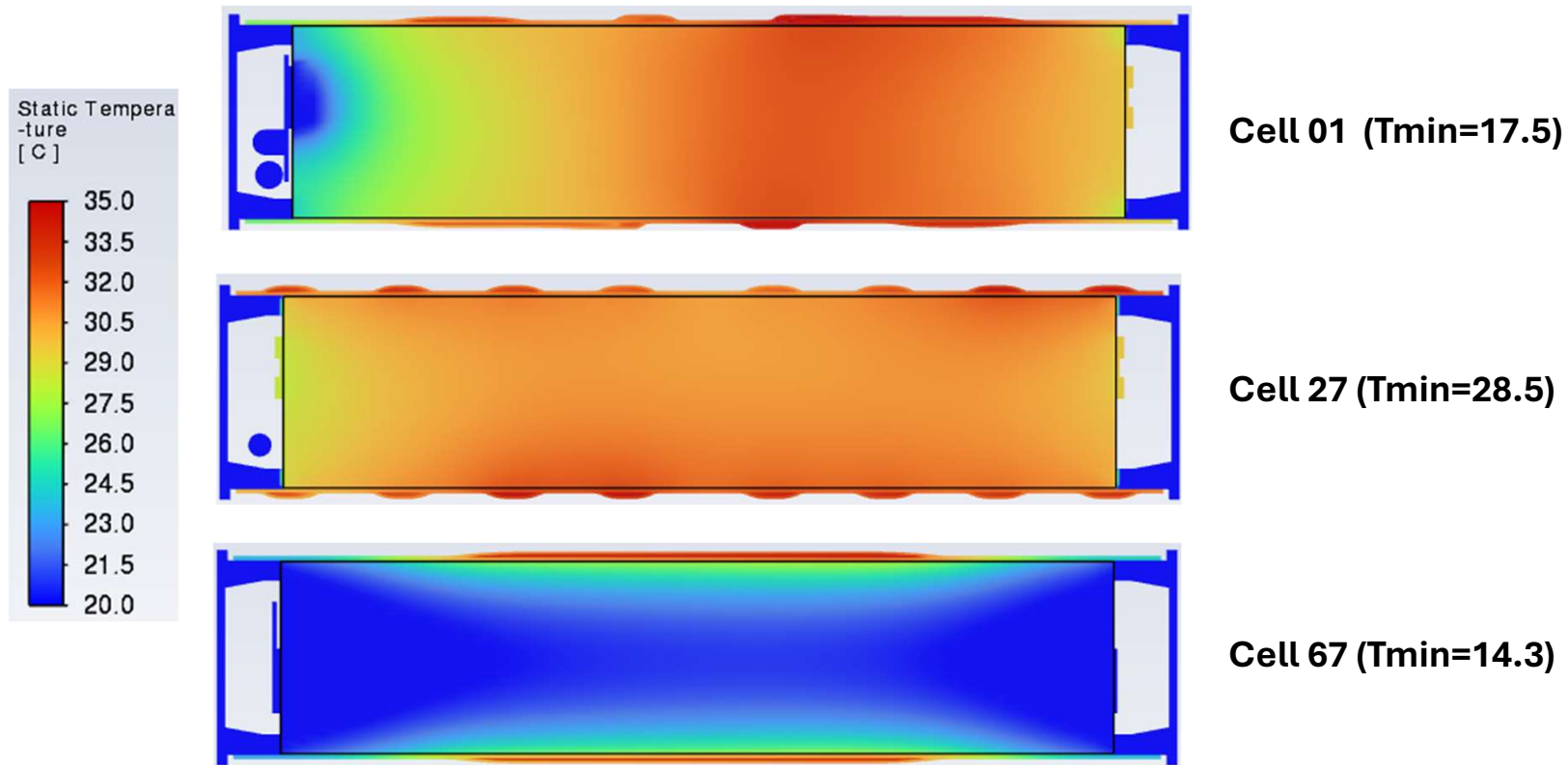


Coolant Inlet temperature & MFR



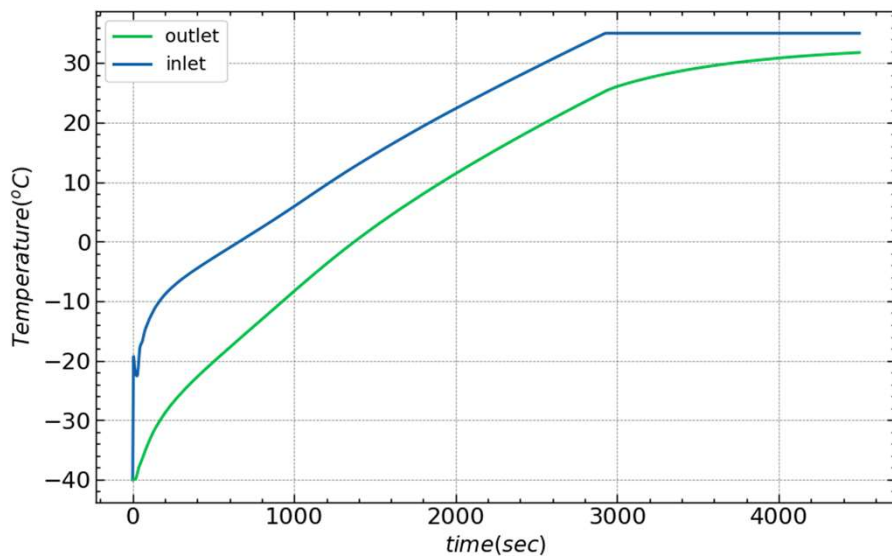
Heat Generation

Cross sectional temperature of Cells @4500s

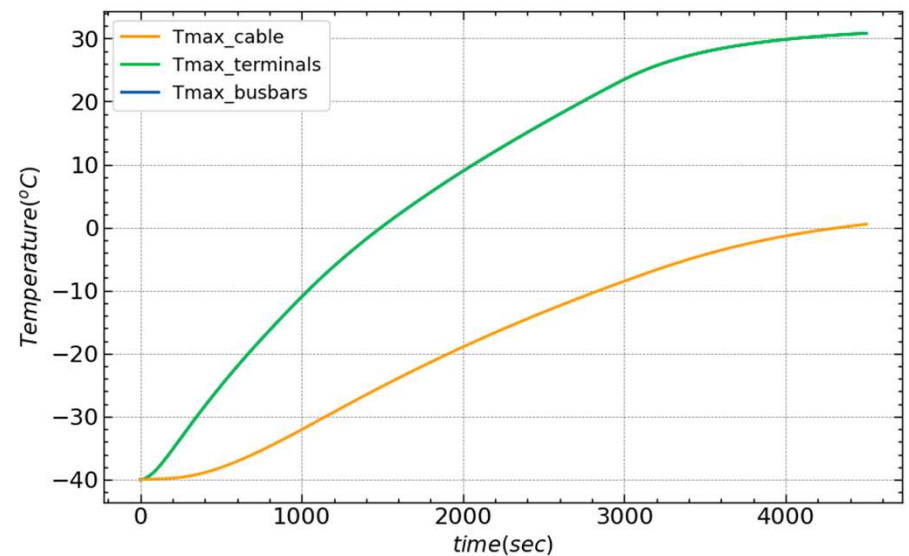


- ❖ Temperature of Cell 27 is higher than that of Cell 01 and Cell 67.
- ❖ Temperature differences in Cell 01 and Cell 67 are more evident than in Cell 27.

Temperature Evolution in Coolant and Components



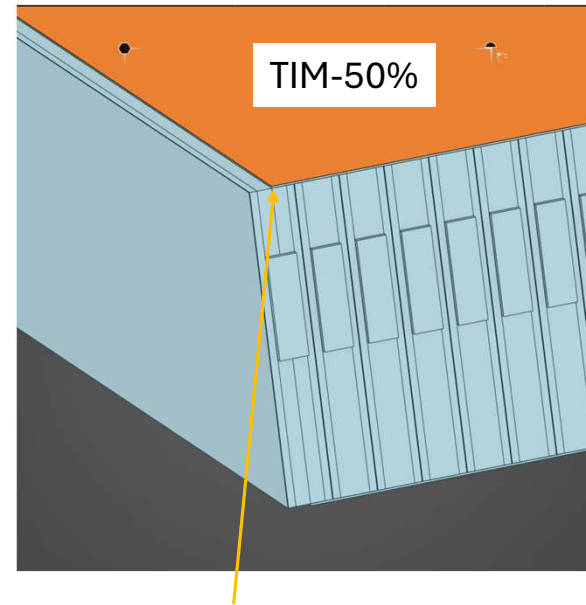
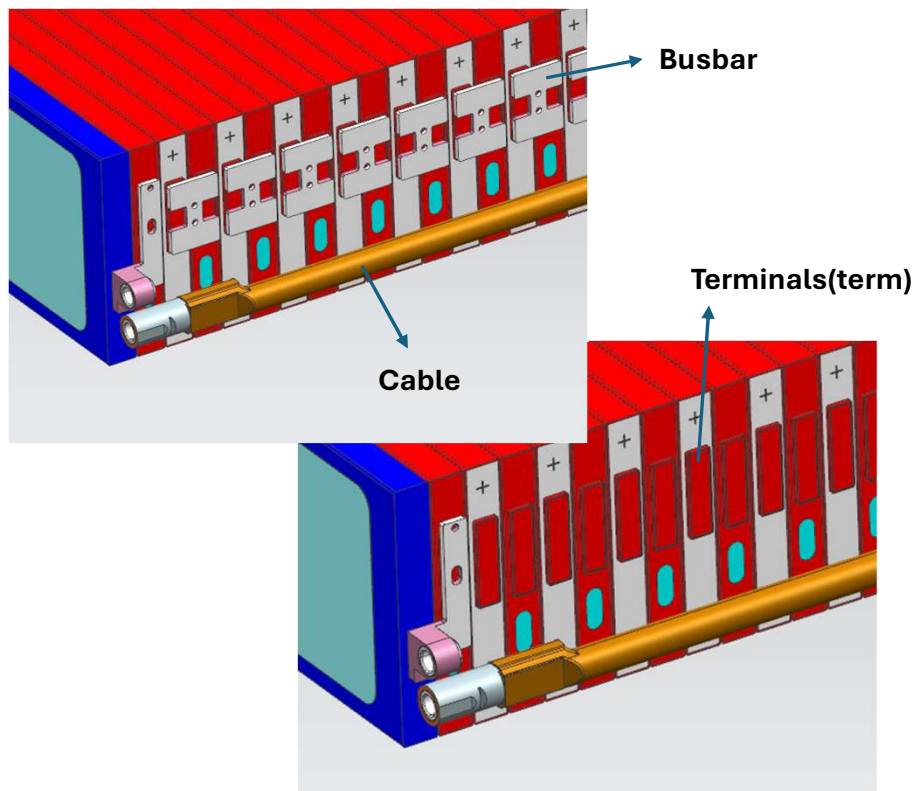
Coolant Temperature Evolution



Max. Temperature in Cable/Terminals/Busbars

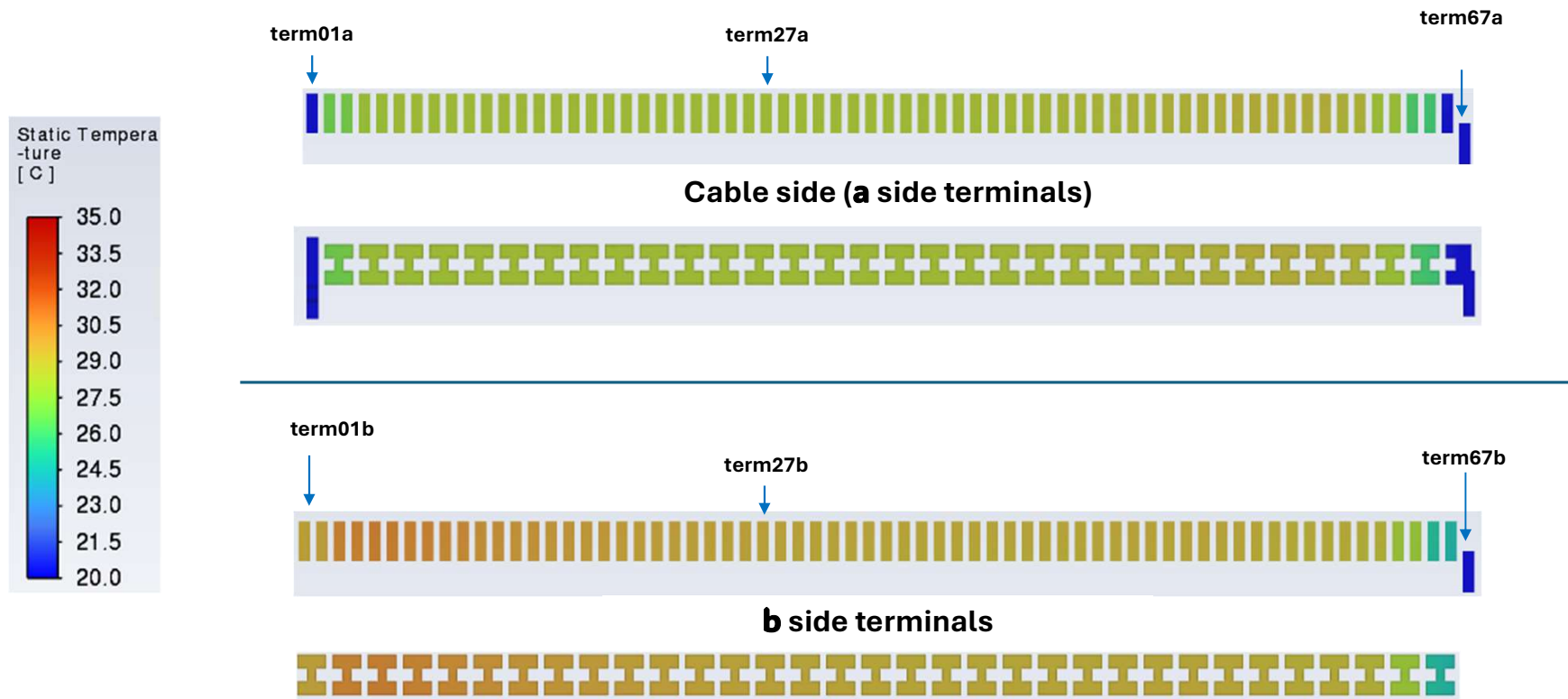
- ❖ Temperature Difference between Coolant inlet and outlet is around 4 °C at the end of cycle.
- ❖ Cable Temperature increases slowly because the heat generation in it is small considering surrounding temperature.

Battery Cells with Terminals and Busbars



TIM covers the top and bottom of cells.
For the two side cells (01, 67), they are covered by 50%

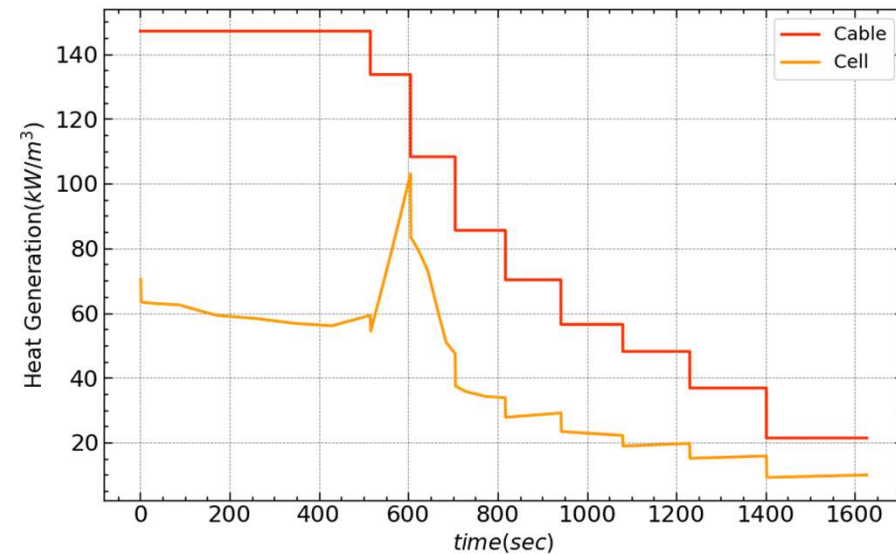
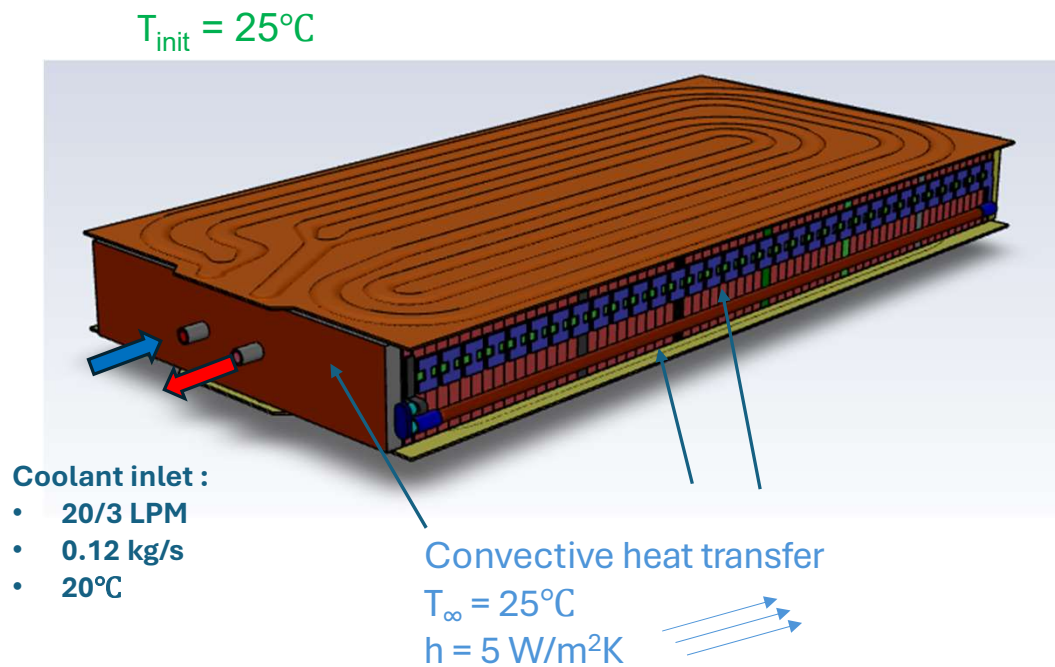
Terminal/Busbar Temperature @ 4500s



❖ Terminals and Busbars contact each other. Their temperature distributions look very similar.

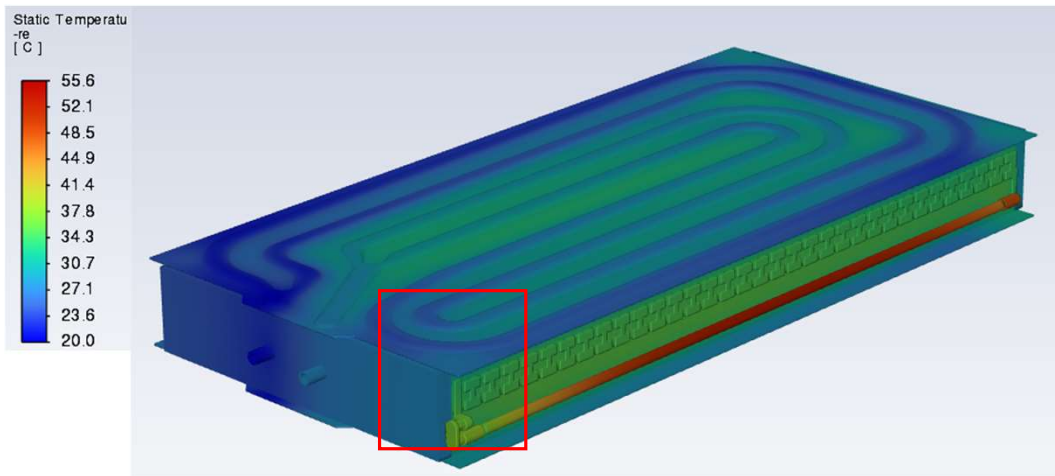
TIM 50%

Boundary Condition & Heat Source

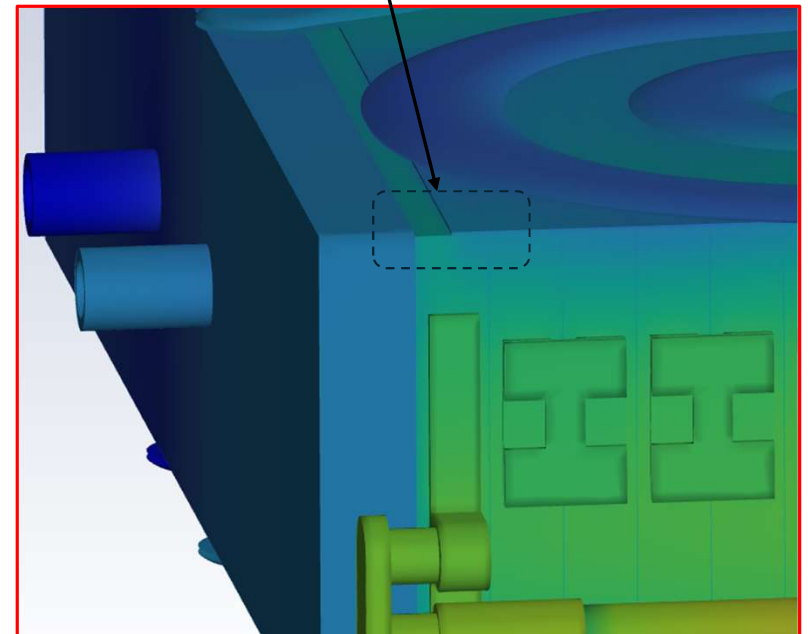


**Heat Cycle on Cells
(Fast_Charging_BOL)**

Temperature Distribution @ 700s

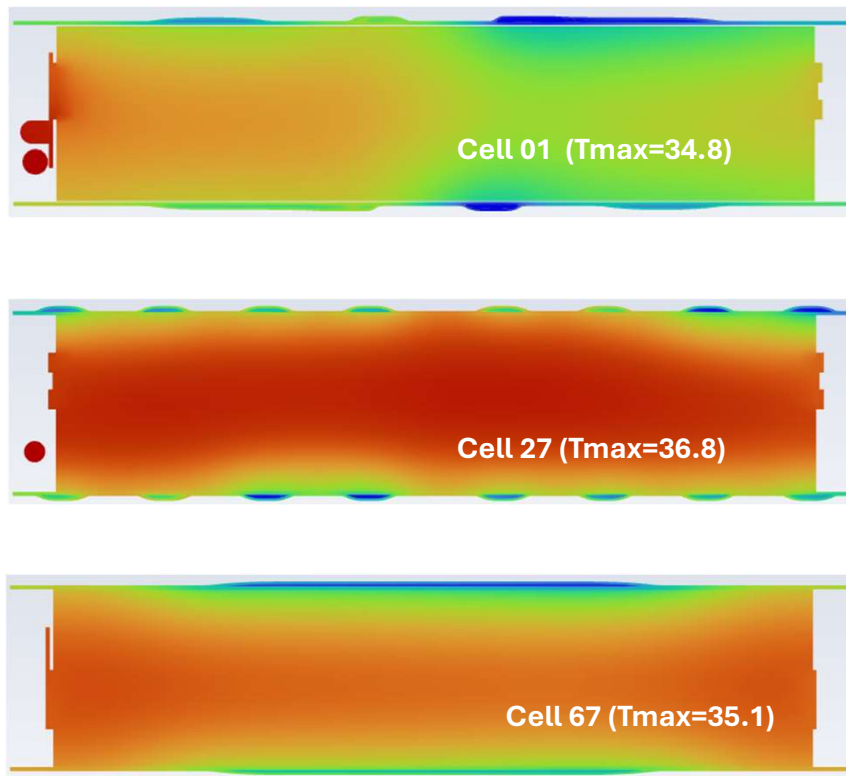


Cell temperature distribution changes from under TIM to without it

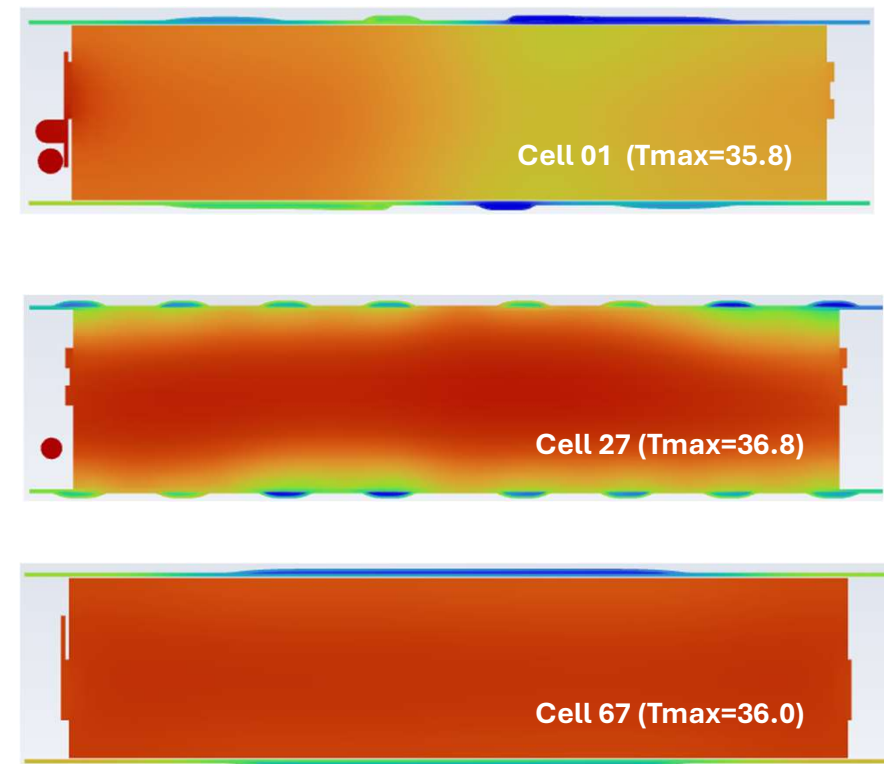


Cross sectional temperature of Cells @700s

50% TIM

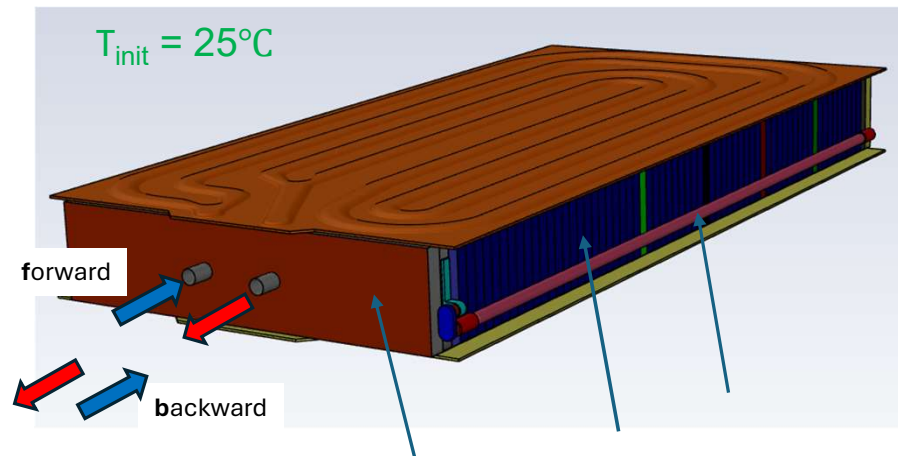


0% TIM



Forward Flow vs. Reverse Flow

Reverse Flow: Boundary Condition & Heat Source



Coolant inlet :

- 20/3 LPM
- 0.12 kg/s
- 20°C

Convective heat transfer

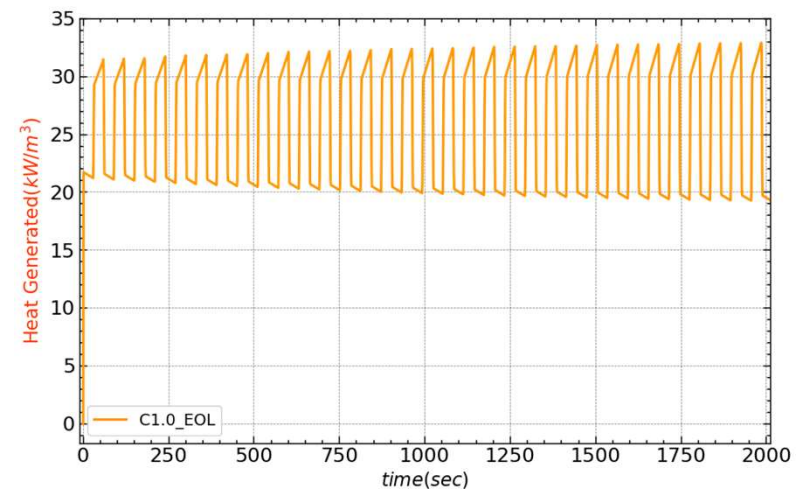
$$T_{\infty} = 25^{\circ}\text{C}$$

$$h = 5 \text{ W/m}^2\text{K}$$



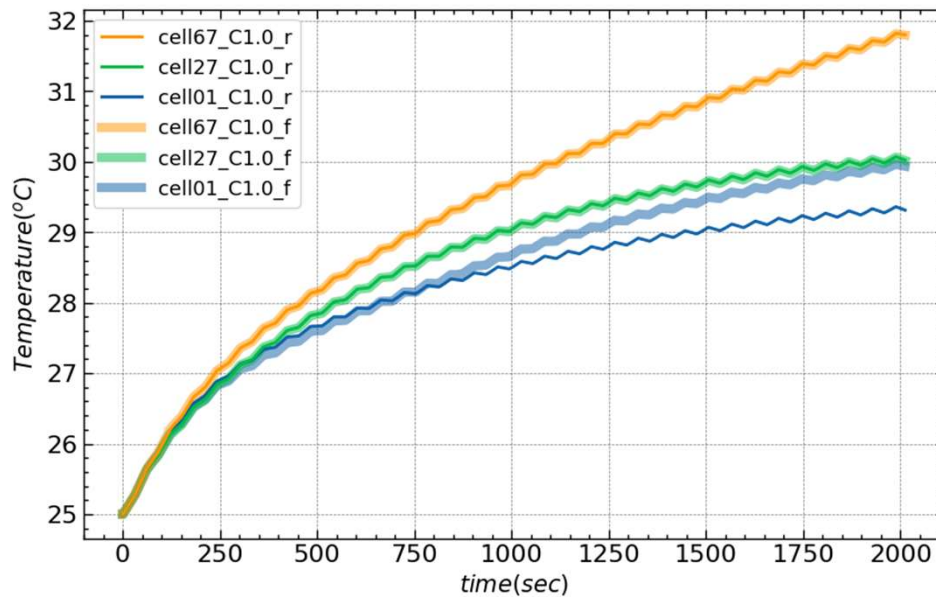
Heat Gen in Cable

- 1.0C : 33420 W/m³

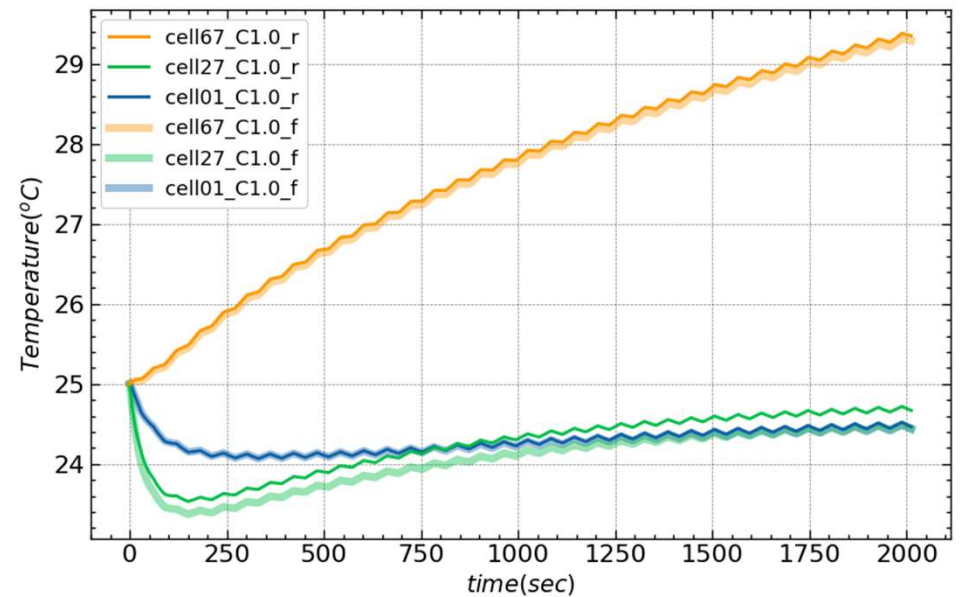


Heat Generation

Comparison between _f(orward) Vs. _r(everse)

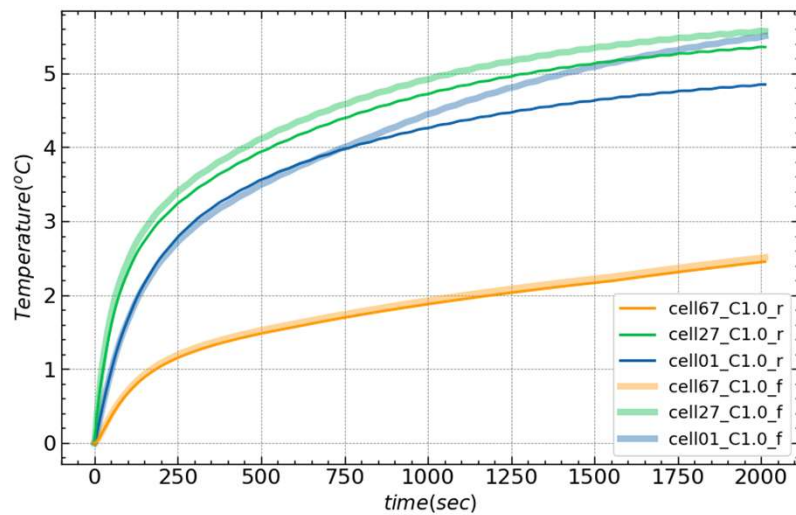


Max. Temperature Evolution

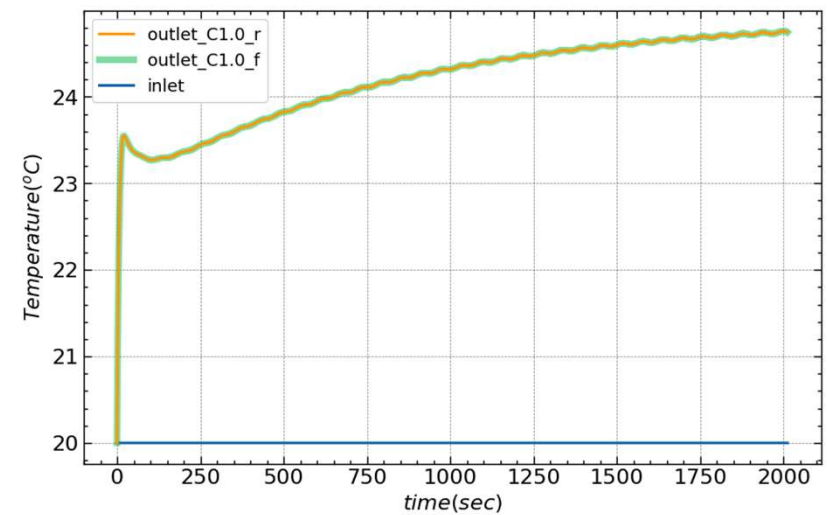


Min. Temperature Evolution

Comparison between _f(orward) Vs. _r(everse)

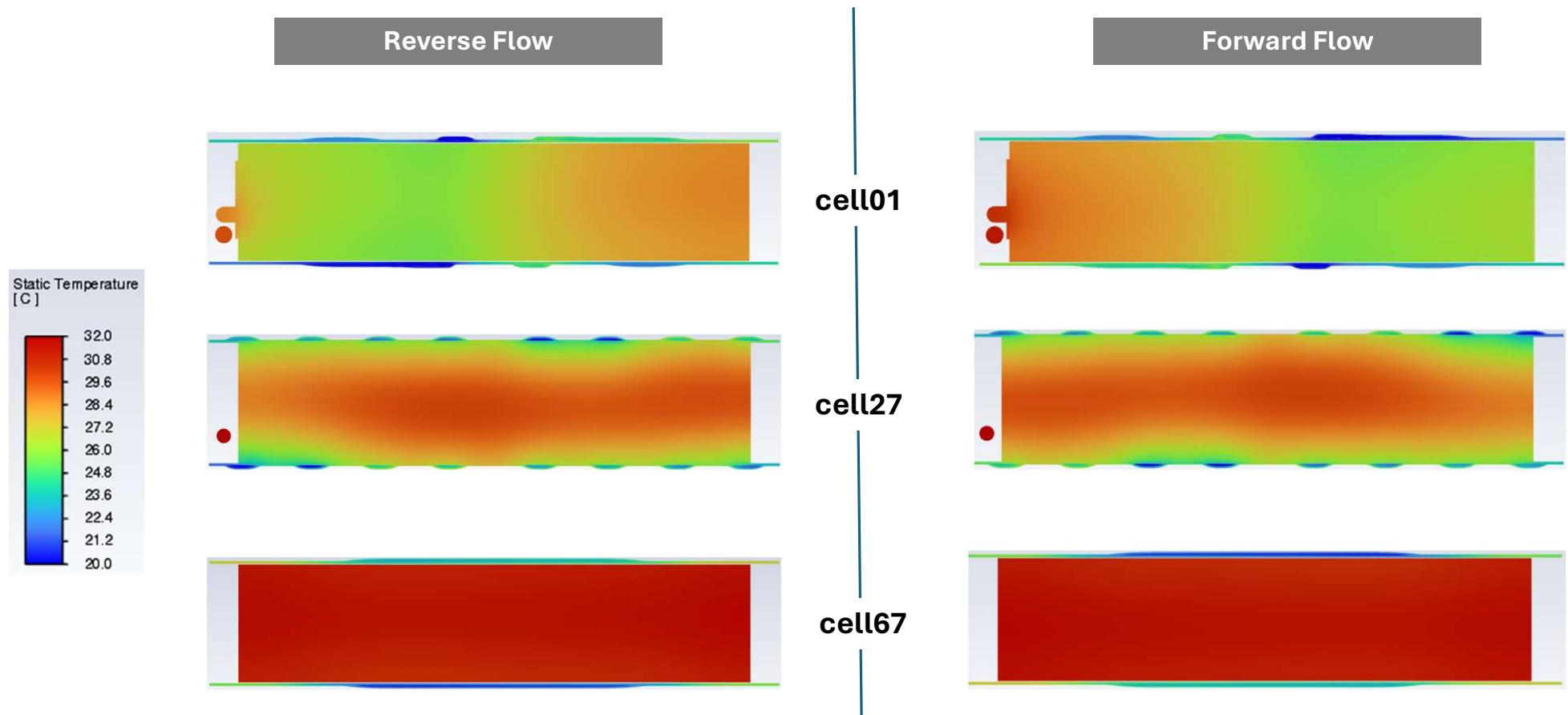


DT (Tmax – Tmin)



Coolant Temperature

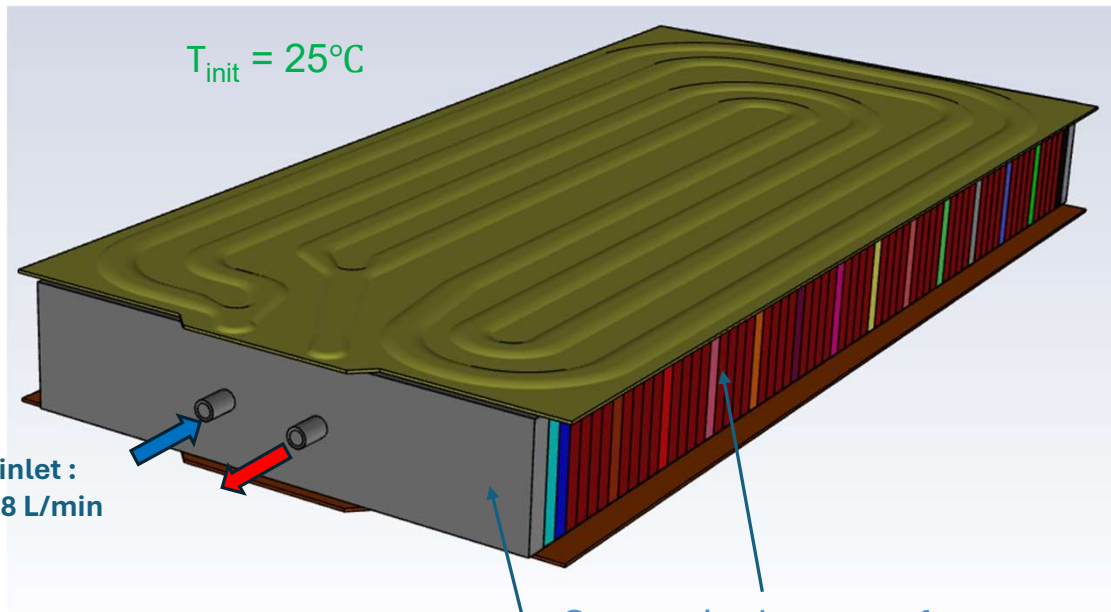
Temperature at Cell cross section @2012sec (1.0C)



Changing Coolant MFR

- **4,6,8lpm**

Boundary Condition & Heat Source

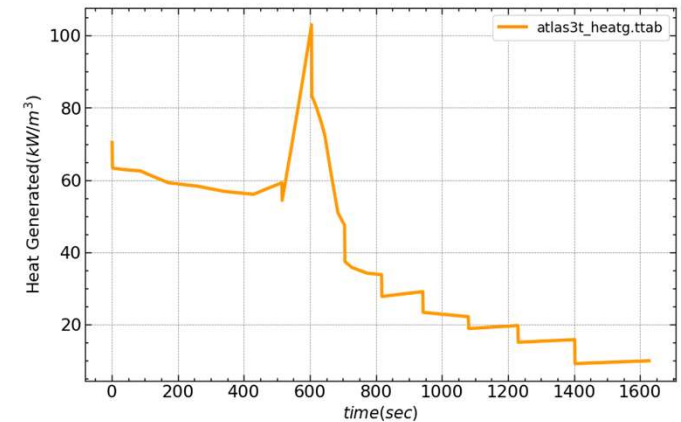


Coolant inlet :

- 4, 6 , 8 L/min
- 20°C

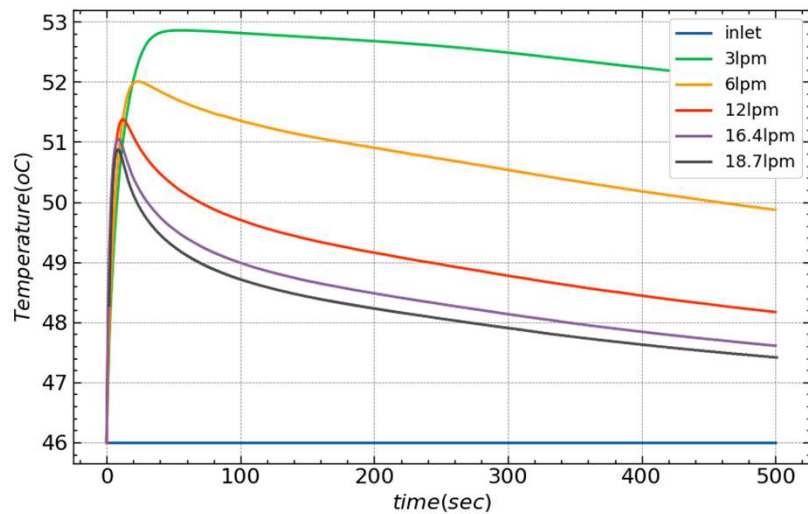
LPM (L/min)	m3/sec	kg/sec
4	6.68E-05	0.07206
6	1.00E-04	0.10809
8	1.34E-04	0.14412

Convective heat transfer
 $T_{\infty} = 25^{\circ}\text{C}$
 $h = 5 \text{ W/m}^2\text{K}$

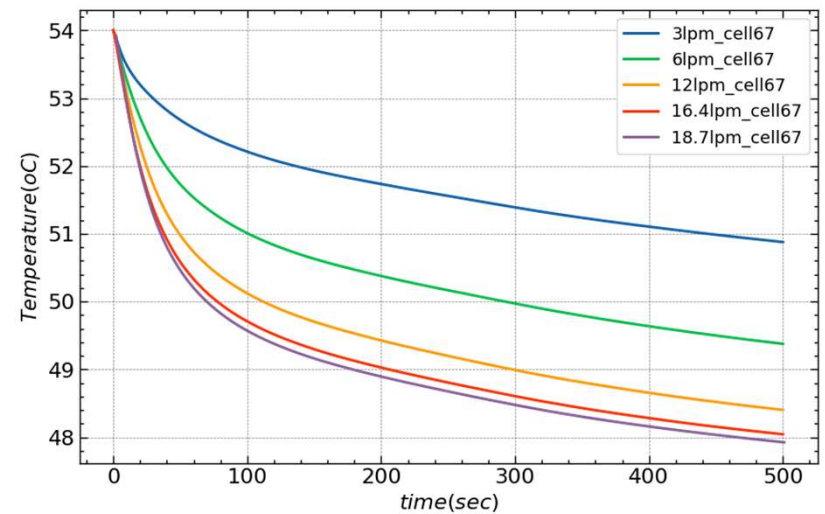


Heat generation while fast charging cells

Temperature Evolution

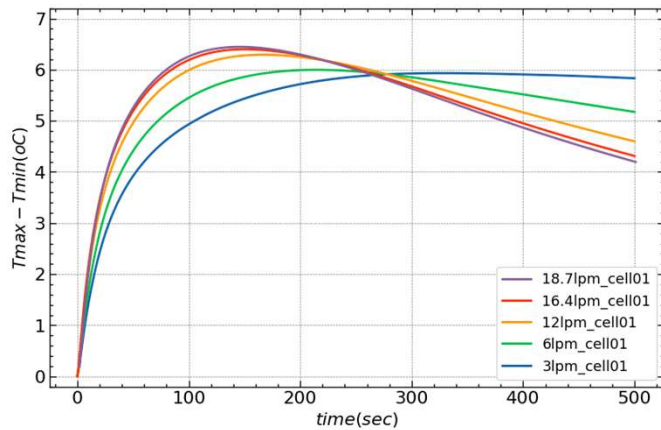


Coolant Temperature

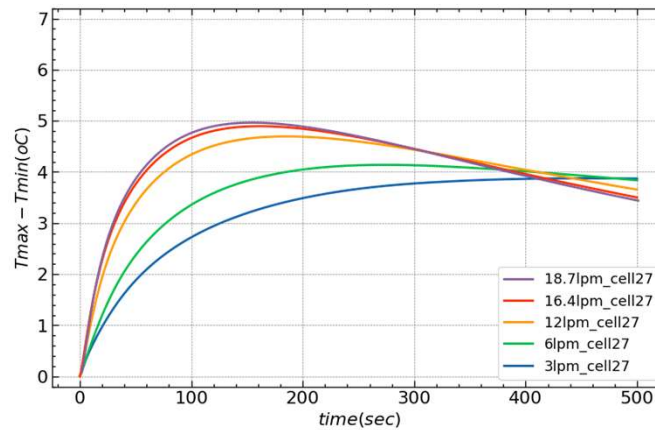


Maximum Temperature of Cells

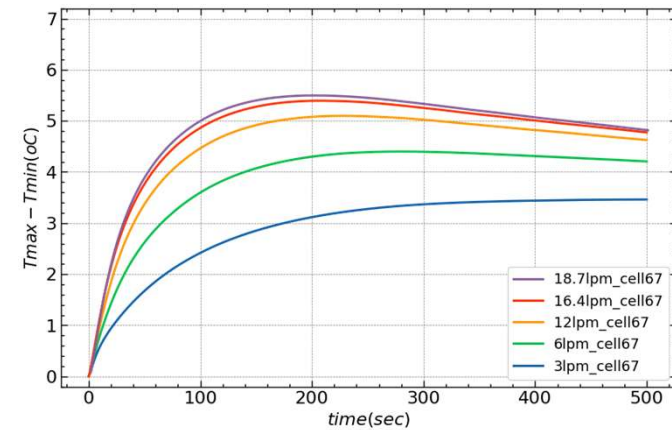
Temperature Difference in Cells



Cell 01



Cell 27



Cell 67